

popquiz adding vectors, subtracting

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:30pm

Quiz Instructions



Question 1 20 pts

. If $\vec{A} = 12\hat{i} - 16\hat{j}$ and $\vec{B} = -24\hat{i} + 10\hat{j}$, what is the magnitude of the vector $\vec{C} = 2\vec{A} - \vec{B}$?

a. 42

b. 22

c. 64

 d. 90

hint: $x_C = 2x_A - x_B$, $y_C = 2y_A - y_B$

In the coordinate system, plot the point (X_C, Y_C) . This is the head of the vector C. Connect origin $(0,0)$ to the head (X_C, Y_C) . Use pythagorean theorem to find the magnitude. You will need the graph for next question.



a



c



d

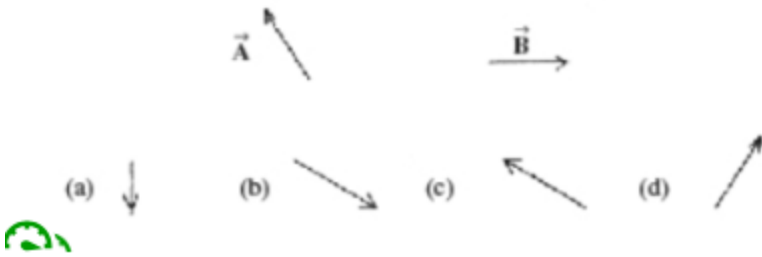


b



Question 2 20 pts

9) Consider two vectors $\vec{A}$ and $\vec{B}$ shown in the figure. The difference $\vec{A} - \vec{B}$ is best illustrated by



hint: Attach A and B by their tails. Go from head of B to head of A (from 2nd term to first for subtraction)

☐

a

☐

d

☐

b

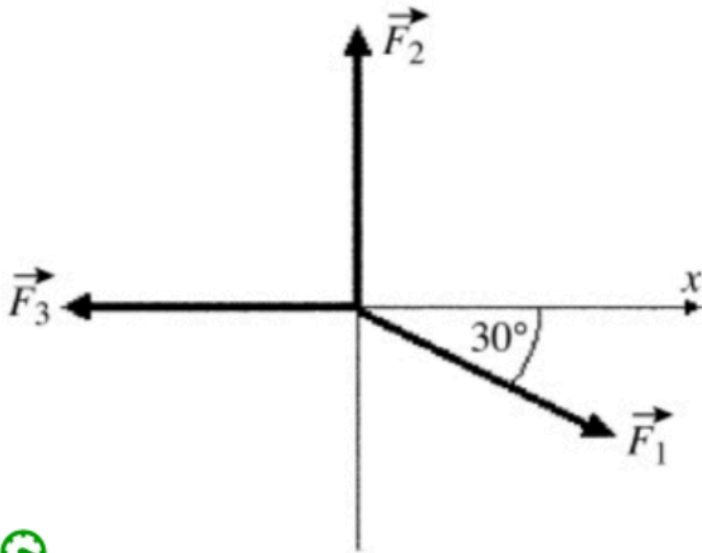
☐

c

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Question 3 20 pts

As shown in the figure, three force vectors act on an object. The magnitudes of the forces as shown in the figure are $F_1 = 80.0 \text{ N}$, $F_2 = 60.0 \text{ N}$, and $F_3 = 40.0 \text{ N}$, where N is the standard SI unit of force. The resultant force acting on the object is given by



- A) 180 N at an angle 60.0° with respect to $+x$ -axis.
- B) 60.0 N at an angle 90.0° with respect to $+x$ -axis.
- C) 20.0 N at an angle 34.3° with respect to $+x$ -axis.
- D) 35.5 N at an angle 34.3° with respect to $+x$ -axis.
- E) 40.0 N at an angle 60.0° with respect to $+x$ -axis.

hint: Add the vectors $F_1 + F_2 + F_3$ Let's call the sum F_{net}

- 1) Find the components of F_1, F_2, F_3 $F_1(x_1, y_1)$; $F_2(x_2, y_2)$; $F_3(x_3, y_3)$
- 2) Find the components of the sum $F_{\text{net}}(x_1+x_2+x_3, y_1+y_2+y_3) = F_{\text{net}}(x, y)$!!! $x_3 < 0$ and $y_1 < 0$!!!!
- 3) In a (X,Y) coordinate system plot the point $F_{\text{net}}(x, y)$. This is the head of the vector. Connect the origin (the tail) to the point $F_{\text{net}}(x, y)$
- 4) Find direction and magnitude. Voila!

For the angle, it is about.

- ☐ B
- ☐ A

☐

D

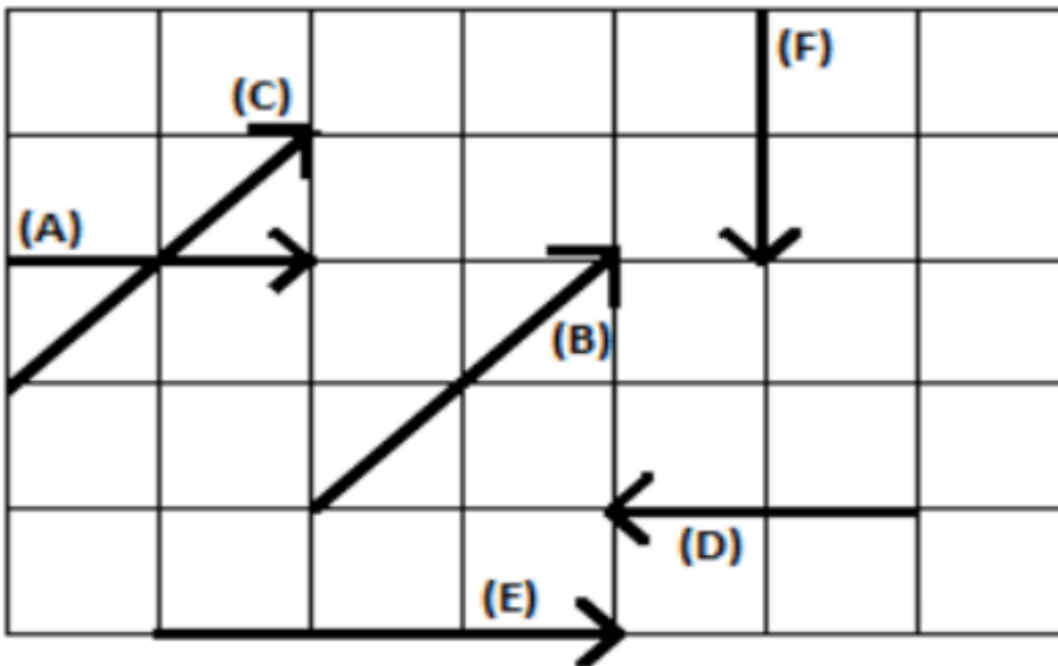
☐

C

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Question 4 20 pts

The sum of vectors A, D and E will be



hint: opposite vectors cancel each other

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A

☐

F

☐

C

☐

D

☐

B

☐

E

☐

Question 5 20 pts

Building on previous question. The sum of vectors B and F

hint: connect the tail of the 2nd vector to the head of the first vector. then connect tail of the first to the head of the second. this is the sum

☐

E

☐

A

☐

C

☐

F

☐

D

Not saved

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